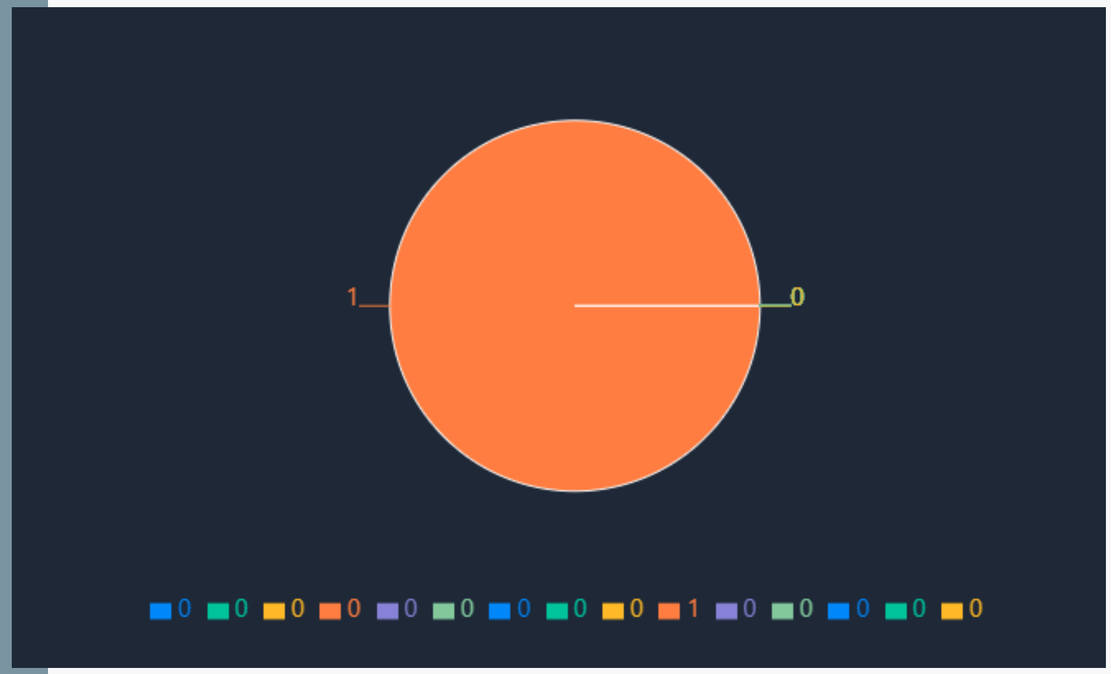
NATICUSdroid (Android Permissions)

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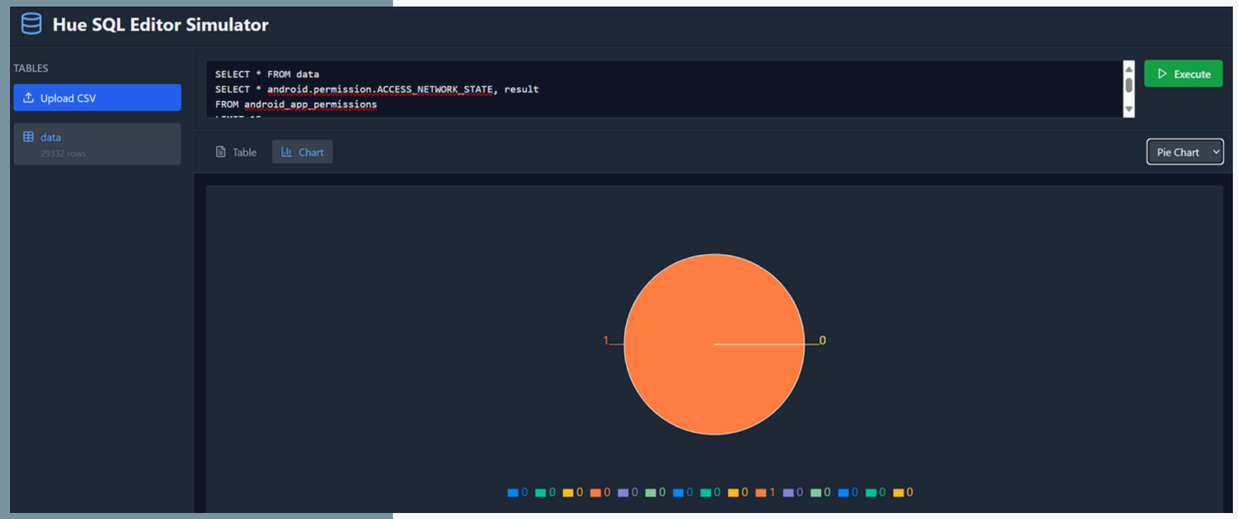
Introduction

- This was for creating a Android malware detection system using latest apps.
- Our project contains permissions extracted from more than 29000 benign & malware Android apps released between 2010-2019.



- SYNTAX:
- SELECT * FROM data
- SELECT * android_permission_read_sms, result
- FROM android_app_permissions
- LIMIT 15;

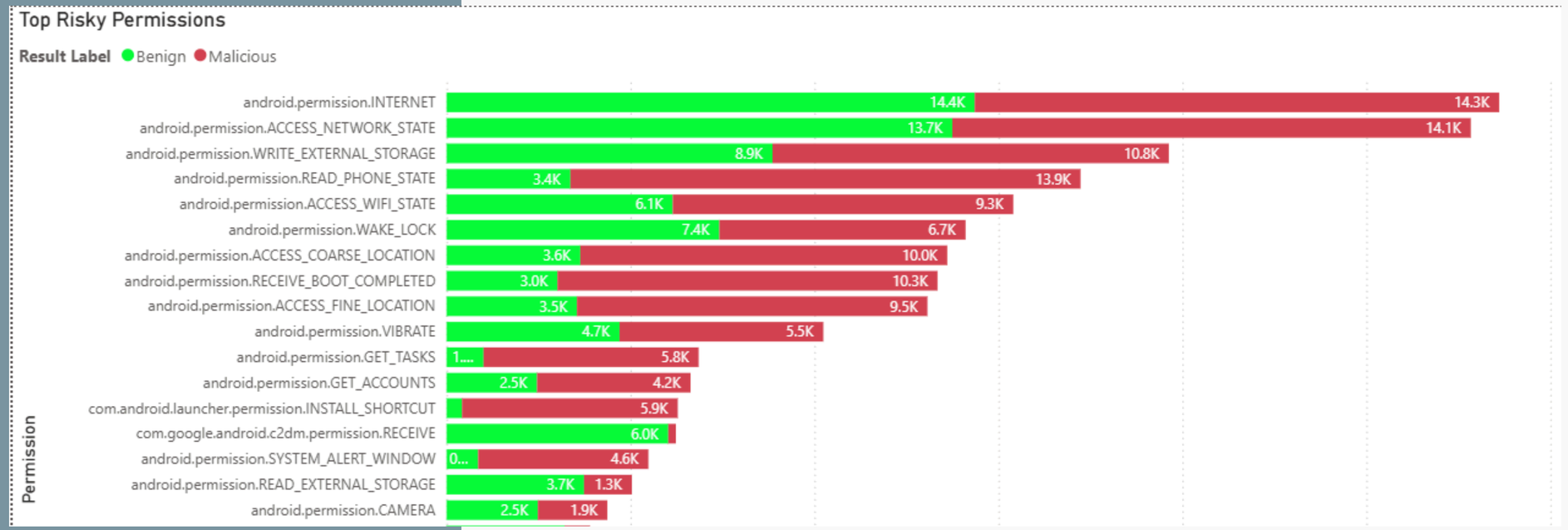
- The SQL query displays 15 apps that request permission to read messages. It shows whether each of these applications is classified as malicious or benign by including the result column.




```
SELECT * FROM data
SELECT *
android.permission.ACCESS_NETWORK_S
TATE, result
FROM android_app_permissions
LIMIT 15;
```

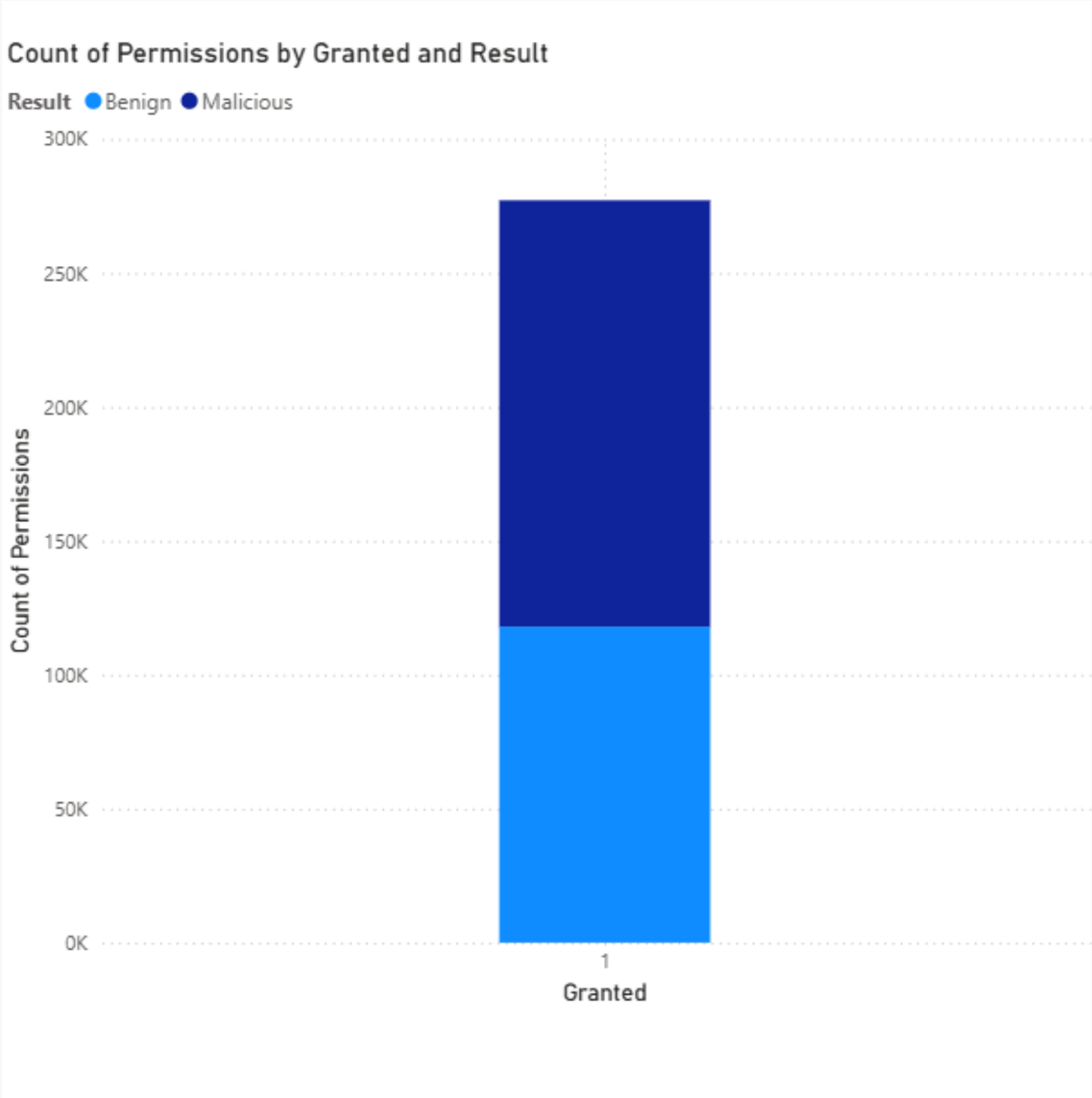
- This query counts the number of Android applications that request permission to access the device’s network state and groups them by malware classification. The result highlights whether network monitoring permissions are more prevalent among malicious or benign applications.

Top Risky Permissions



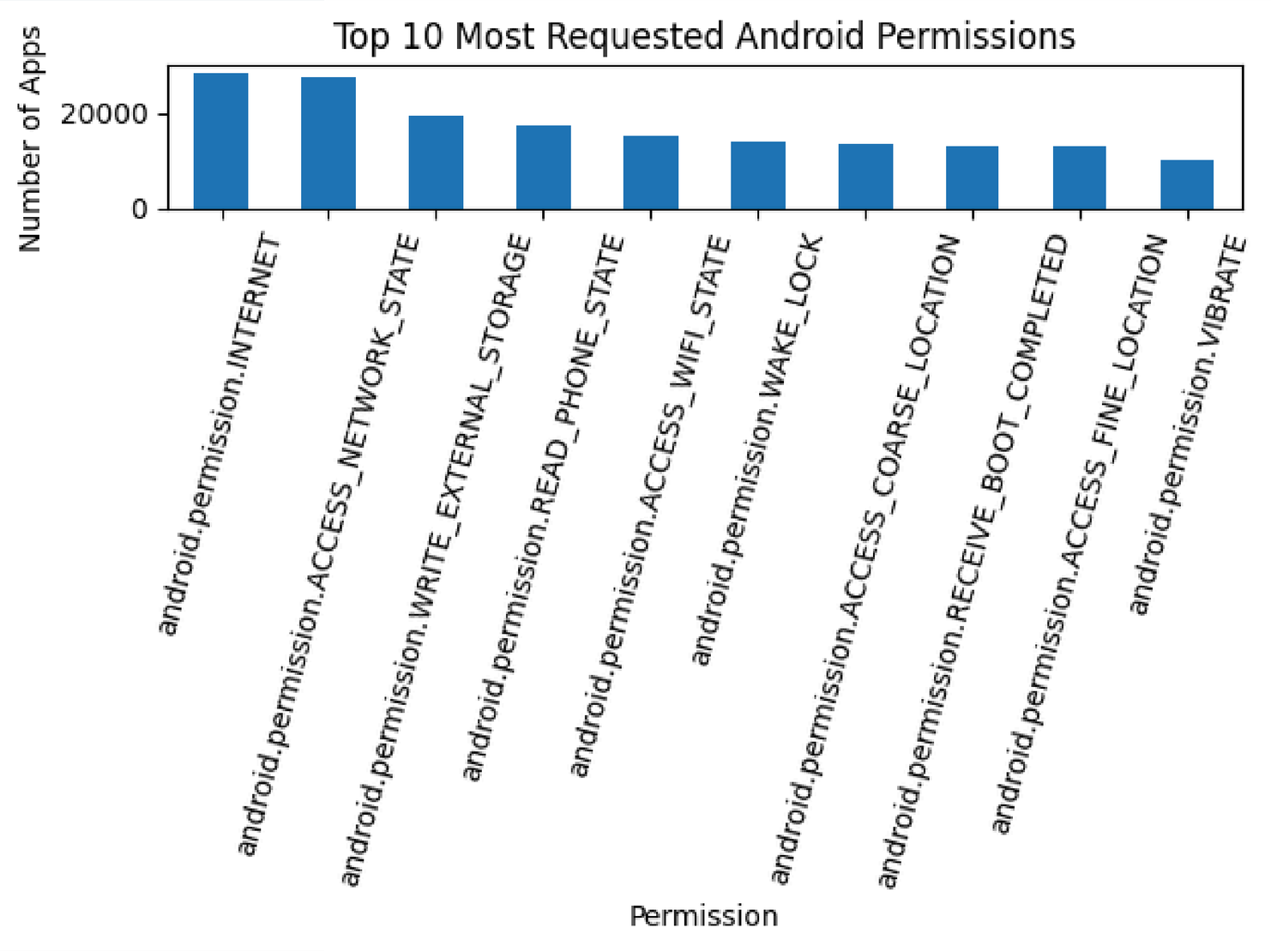
- Permissions related to **phone state, location, and boot receive** show the **highest malicious usage ratios**, making them the "riskiest" permissions to grant.

- This analysis helps identify which permissions are most commonly abused by malicious apps versus those that are generally safe for legitimate applications.



- This graph shows the frequency of app permissions in Benign and Malicious apps, highlighting permissions like [Permission A] and [Permission B] that are more common in Malicious apps, while others like [Permission C] appear mostly in Benign apps.

- Some permissions appear in both but at different frequencies, suggesting potential misuse in Malicious apps and helping identify risk indicators for monitoring and security assessments.



- These permissions are requested by the majority of the apps. This indicates that most Android applications rely on internet connectivity for their features.

- The graph shows that most Android apps request internet and network-related permissions, while fewer apps request more sensitive permissions like location and phone state, highlighting common usage patterns and potential privacy risks across a large app ecosystem.

Conclusion

The analysis shows that permissions related to phone state, location access, and boot reception have the highest association with malicious applications, making them the riskiest permissions to grant. While internet and network-related permissions are commonly requested by both benign and malicious apps due to functional requirements, differences in their usage frequency suggest potential misuse by malware. Some permissions appear in both categories but at varying levels, indicating their value as indicators for identifying suspicious behavior.

Aggregate queries caused instability in Hue, so Power BI was used to visualize the results more effectively, aligning with the project's primary objective of data visualization. The permission-based analysis combined with clear visual analytics provides valuable insight into Android malware behavior and supports better security assessment and monitoring.

References:

- https://archive.ics.uci.edu/dataset/722/naticusdroid+android+permissions+dataset

Thank you